

Pavan Kumar Balija

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PROFILE OVERVIEW

- Software Developer with **4** years of experience in the **Adaptive AUTOSAR platform** and **Modern C++ (C++11/14/17) development**.
- Proficient in designing and developing **Basic Software Components** in compliance with **Adaptive AUTOSAR specifications**.
- Successfully **designed and implemented sample applications** to validate AUTOSAR functionalities and features.
- Hands-on experience with **Adaptive AUTOSAR configuration tools** and integration workflows.
- Involved in the **development and functional testing** of key Adaptive AUTOSAR modules such as **Cryptography, ara::core, Diagnostics**.
- Strong technical knowledge of Adaptive AUTOSAR components, including **COM, SM (State Management), DIAG (Diagnostics), CRYPTO(Cryptography), and EXEC (Execution Manager), etc...**
- Experience in **integrating the Adaptive AUTOSAR stack on QEMU virtual platforms**, ensuring end-to-end stack functionality and validation.
- Well-versed in **software development methodologies** including **Agile, V-Model (SDLC)**, and coding standards such as **AUTOSAR C++14 (MISRA compliant)**.
- Managing **customer queries**, involving in issues and **bug fix of the products**.
- Experience in working with continuous integration and continuous deployment (CI/CD) environment **using Jenkins, Jira, Bitbucket, Git, Jfrog Artifactory, Confluence, Docker, VS Code, etc...**
- Demonstrated ability to **analyse customer requirements** and translate them into functional, modular, and scalable software components.
- A **proactive team player** with strong **analytical, problem-solving, and interpersonal skills**, committed to delivering high-quality embedded solutions in automotive domains.

TECHNICAL SKILLS:

Programming Languages:	Modern C++, C++ (11/14/17), STL, Multithreading and C - Language.
Development Tools	Visual Studio Code, CANoe, GitHub, Vector Cast Tool, Davinci developer Adaptive tool, Atlassian Tools (Bitbucket, Jira), Code Beamer (ALM), Axivion tool, AWS EC2 Instance, Shell Scripting, CMake, JSON, Yocto, QEMU.
Operating System	Linux (Ubuntu), Windows, QNX.
Domain Expertise	Adaptive AUTOSAR, Classic AUTOSAR and ADAS.
Software Methodologies	Agile Methodology, V-Model (SDLC), AUTOSAR C++14 (MISRA compliant)
Protocols & System Programming	TCP/IP, UDP, SOME/IP, IPC Mechanisms.

EDUCATION:

Qualification	Institution	Board/University	Passing year	Marks In %
B.Tech (ECE)	SMEISGI	JNTUA	2020	70.68
Class XII	GOVERNMENT JUNIOR COLLEGE	BOARD OF INTERMEDIATE, AP	2016	87.9
Class X	S R K MPL BOYES HIGH SCHOOL	BOARD OF SECONDARY, AP	2014	80

PROFESSIONAL EXPERIENCE:

- **KPIT Technologies Ltd** (Oct/24/2024 - Present) - Sr Software Engineer - Adaptive AUTOSAR
- **AVIN Systems Pvt Ltd** (July/15/2021 - Oct/23/2024) - Sr Software Engineer - Adaptive AUTOSAR

AWARDS:

- **Thankyou Award**, AVIN Systems Pvt Ltd, Dec 2023.
Description: Completing activities with minimal guidance.
- **Excellence Award**, AVIN Systems Pvt Ltd, Jan 2024.
Description: Having an ability to maneuver through multiple domains with commitment & dedication.
- **Above & Beyonders Award**, AVIN Systems Pvt Ltd, Jan 2024.
Description: In recognition and appreciation of outstanding performance and contribution to the project.

PROJECTS UNDERTAKEN:

PROJECT 1:

Project Name	Integration and Development of Diagnostic Stack
Project Description	Bug fixing and Implementation of overall DIAGNOSTIC according to AUTOSAR specification and customer requirements.
Contribution	My responsibilities included: ➤ Analysed customer requirements based on AUTOSAR Diagnostic specification doc. ➤ Identified and fixed all defects in the Diagnostic stack and developed code as per updated requirement. ➤ Performed unit testing of the complete Diagnostic stack using the VectorCAST tool. ➤ Resolved all static code violations in the Diagnostic stack, ensuring compliance with coding standards. ➤ Participated in ASPICE activities, contributing to process improvement and quality assurance.
Tools Used	Visual Studio Code, Ubuntu, JIRA, CANoe, GitHub, Vector Cast Tool, Davinci tools.

PROJECT 2:

Project Name	ara::core Module Development on Adaptive AUTOSAR Platform
Project Description	Implementation and integration of overall Adaptive AUTOSAR FC's according to AUTOSAR specification and customer requirements. Adaptive ara::core is one of the module used to provide the advance C++ libraries to all AUTOSAR modules.
Contribution	My responsibilities included: ➤ Analysed requirements from the AUTOSAR Adaptive ara::core SWS (20-11) specification doc. ➤ Integrated the ICEORXY C++ feature into the ara::core module and modified code based on requirements. ➤ Performed unit testing of the ara::core module in alignment with development code. ➤ Conducted integration testing to validate overall module functionality and component interactions. ➤ Created sequence diagram scripts and generated diagrams using PlantUML for architectural visualization. ➤ Resolved static code violations in the Adaptive Core codebase to maintain coding standard compliance.
Tools Used	Visual Studio Code, Atlassian Tools (Bitbucket, Jira), Code Beamer (ALM), GitHub, Axivion tool, JIRA.

PROJECT 3:

Project Name	Cryptography Module Development on Adaptive AUTOSAR Platform
Project Description	Implementation and integration of overall Adaptive AUTOSAR FC's according to AUTOSAR specification and customer requirement. Cryptography is one of the modules used to provide cybersecurity of the applications in the vehicle from the hacker attacks and to verify the authenticity and integrity of the data.
Contribution	My responsibilities included: ➤ Analysed requirements from the AUTOSAR Cryptography SWS (20-11) specification document. ➤ Implemented Cryptography Functional Cluster (FC) based on the specifications defined in the AUTOSAR_SWS_Cryptography document. ➤ Configured Cryptography interfaces using the ASTRA configuration tool (Davinci developer Adaptive). ➤ Developed basic test applications (such as qt-app and qt-tester) to invoke and validate Cryptography APIs. ➤ Configure the SOME/IP (P-port and R-port) services on Davinci developer Adaptive. ➤ Performed functional testing of the Cryptography module in line with the development code. ➤ Created sequence diagram scripts and generated diagrams using PlantUML for system behaviour visualization. ➤ Resolved static code violations in the Cryptography development code.
Tools Used	VS Code, Davinci developer Adaptive tool, Bitbucket, Jira, Code Beamer, GitHub, Axivion tool, CANoe.

PROJECT 4(Current):

Project Name	Integration and Validation of Adaptive AUTOSAR Stack on Virtual Platforms (QEMU)
Project Description	Involved in Integration the Adaptive AUTOSAR stack on QEMU environments. The project required ensuring that the stack worked as expected in these environments, identifying any issues, providing solutions, and documenting the changes encountered. This also included validation of the stack's performance, functionality, and integration with the virtual platforms.
Contribution	My responsibilities included: ➤ Integrated the Adaptive AUTOSAR stack into QEMU virtual environments. ➤ Ensured proper functionality and performance of the stack within virtualized platforms. ➤ Identified defects, troubleshoot issues, and implemented effective solutions. ➤ Utilized Yocto to build the Adaptive AUTOSAR stack and generate the required ext4 file and kernel image. ➤ Deployed the generated images into the QEMU virtual environment for validation and testing. ➤ Developed communication between the QEMU platform and a Linux platform to send and receive data using the SOME/IP protocol, including code implementation and modifications as per requirements. ➤ Documented challenges encountered and detailed the solutions applied during the integration and testing phases.
Tools Used	Visual Studio Code, AWS EC2 Instance, Ubuntu, Yocto, QEMU.

STRENGTHS:

- Positive Learning Attitude.
- Dedication and Flexible to work.
- Team Worker.
- Good listener and communicator.
- Quick Learner.

DECLARATION:

I do hereby declare that the particulars of information and facts stated here in above are true, correct and complete to the best of knowledge and belief.

Yours sincerely
Baliya Pavan Kumar

